

What is pose detection?

This refers to the process of an AI detecting the position and posture of a human just from an image or a video. This is done by detecting where the body points like elbows, shoulders, knees and ankles are, basically the joints of a human body. And converts these body points into coordinates with x and y to determine which body point is where and how far from each other. Hence creating the shape of the person.

- The AI has to be trained on millions of human body photos consisting of different poses to understand and give more accurate results.
- Basically, in any position, the ai has to be able to detect and measure the body lengths of a human and hence removing the need for a measuring scale. This is very important for the Raritone Application as without it virtual try-on feature(The feature the entire application is based on) directly fails to detect a human body itself.

For these the assigned tools and frameworks are perfect because they own have a different function but when worked together can achieve pose detection which would help in the development of the raritone application.

1) **OpenCV**: A computer vision library used to read images and videos from a camera. It helps the computer to see the images, by converting them into a number of pixels.

2) **Tensorflow**: A framework used to build Machine Learning models used to predict something. In this case, its perfect to predict which size fits a person best according to their body shape.

3) **Mediapipe**: This is a framework used for functions like pose detection and real time tracking. This uses body point like I mentioned earlier to create a body model through which the body shape is predicted by the tensorflow.

How can AI help fashion Shopping?

AI helps fashion shopping by making the experience more personalized and interactive. Before AI, online shopping gave us no way to test out the dresses or even suggested more similar fashion that we would like. AI helps in fashion shopping in many ways like Virtual try-on, personal recommendation, Trend prediction, easy searching and many more.

- 1) Through training with different datasets that include usual customer activity, previous reviews and customer buying pattern, personal recommendation is achieved.
- 2) AI also helps with features like Virtual try-on which basically refers to being able to try clothes before buying to see how you in the clothes and which size is perfect for you. This also improves confidence of the customer to buy a selected product.
- 3) With pose detection, using ai tools customers can see which size suits them perfectly which the AI will select by itself as it is trained to determine to choose the best clothing size according to the body size of the customer. This reduces product returns.
- 4) Applications like Google Lens use AI to find required products using an image and giving the exact link that the customer can use to buy that particular product. This helps the customer to easily and quickly find any desired product hence improving customer satisfaction.
- 5) Fashion companies also use AI to predict fashion trends on different social media and also in the fashion industry itself so they can create styles that are popular currently hence helping the customer to choose from a wider variety of collection.

Just like in every other sector, AI has helped in improving the fashion sector also. Not only for the customers but also for the sellers. So it is important to understand how to implement these ai tools in order to improve sales, customer experience, and to improve the fashion industry overall.